

36. Develop a document Question Answering system using Retrieval-Augmented Generation (RAG).

Code:

```
from sentence_transformers import SentenceTransformer
from transformers import pipeline
import faiss
import numpy as np
```

```
# -----
```

```
# 1. Load AI models
```

```
# -----
```

```
print("Loading embedding model...")
```

```
embedding_model = SentenceTransformer(
    "all-MiniLM-L6-v2"
)
```

```
print("Loading question answering model...")
```

```
qa_model = pipeline(
    "text2text-generation",
    model="google/flan-t5-small"
)
```

```
# -----
```

```
# 2. Document
```

```
# -----
```

```
document = ""
```

Artificial Intelligence is a branch of computer science
that focuses on creating intelligent machines.

Machine Learning is a part of Artificial Intelligence.

It allows computers to learn patterns from data and make
predictions or decisions without being explicitly programmed.

Deep Learning is a type of Machine Learning that uses
artificial neural networks with multiple layers.

Natural Language Processing, also called NLP, is a field
of Artificial Intelligence that helps computers understand,
process and generate human language.

Computer Vision is another field of Artificial Intelligence
that enables computers to understand images and videos.

```
""
```

```
# -----
```

```
# 3. Split document into chunks
```

```
# -----
```

```
chunks = [
```

```
    "Artificial Intelligence is a branch of computer science "
```

```
    "that focuses on creating intelligent machines.",
```

```
    "Machine Learning is a part of Artificial Intelligence. "
```

```
    "It allows computers to learn patterns from data and make "
```

"predictions or decisions without being explicitly programmed.",

"Deep Learning is a type of Machine Learning that uses "

"artificial neural networks with multiple layers.",

"Natural Language Processing, also called NLP, is a field "

"of Artificial Intelligence that helps computers understand, "

"process and generate human language.",

"Computer Vision is another field of Artificial Intelligence "

"that enables computers to understand images and videos."

]

```
# -----
```

```
# 4. Generate embeddings
```

```
# -----
```

```
print("Creating document embeddings...")
```

```
embeddings = embedding_model.encode(chunks)
```

```
embeddings = np.array(
```

```
    embeddings
```

```
).astype("float32")
```

```
# -----
```

```
# 5. Create FAISS vector database
```

```
# -----
```

```
dimension = embeddings.shape[1]
```

```
index = faiss.IndexFlatL2(dimension)
```

```
index.add(embeddings)
```

```
print("Document stored successfully.")
```

```
# -----
```

```
# 6. Get user question
```

```
# -----
```

```
question = input(
```

```
    "\nEnter your question: "
```

```
)
```

```
# -----
```

```
# 7. Generate question embedding
```

```
# -----
```

```
question_embedding = embedding_model.encode(
```

```
    [question]
```

```
)
```

```
question_embedding = np.array(
```

```
    question_embedding
```

```
).astype("float32")
```

```
# -----
```

```
# 8. Retrieve relevant document chunk
```

```
# -----
```

```
distance, result = index.search(  
    question_embedding,  
    2  
)
```

```
# Retrieve top 2 chunks
```

```
retrieved_chunks = []
```

```
for i in result[0]:  
    retrieved_chunks.append(chunks[i])
```

```
context = " ".join(retrieved_chunks)
```

```
# -----
```

```
# 9. Generate answer using retrieved context
```

```
# -----
```

```
prompt = ""
```

```
Answer the question using only the given context.
```

```
Context:
```

```
{
```

```
Question:
```

```
{
```

Answer:

```
"".format(context, question)
```

```
response = qa_model(  
    prompt,  
    max_new_tokens=100  
)
```

```
answer = response[0]["generated_text"]
```

```
# -----
```

```
# 10. Display results
```

```
# -----
```

```
print("\n-----")
```

```
print("RAG QUESTION ANSWERING SYSTEM")
```

```
print("-----")
```

```
print("\nRetrieved Context:")
```

```
print(context)
```

```
print("\nQuestion:")
```

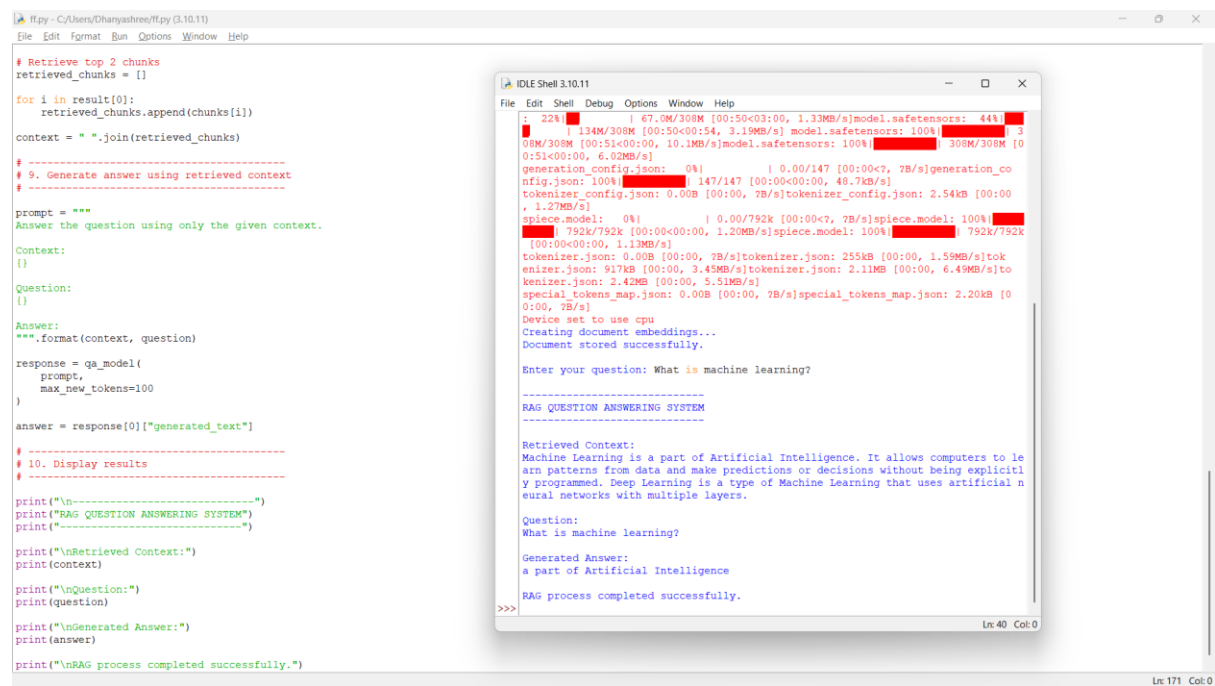
```
print(question)
```

```
print("\nGenerated Answer:")
```

```
print(answer)
```

```
print("\nRAG process completed successfully.")
```

output:



The image shows a Jupyter Notebook interface with a code cell and its corresponding terminal output. The code cell contains Python code for a RAG (Retrieval-Augmented Generation) system. The terminal output shows the execution of the code, including the retrieval of context, the generation of an answer, and the final output of the RAG process.

```
# Retrieve top 2 chunks
retrieved_chunks = []

for i in result[0]:
    retrieved_chunks.append(chunks[i])

context = " ".join(retrieved_chunks)

# 9. Generate answer using retrieved context
# -----
prompt = """
Answer the question using only the given context.
Context:
{}
Question:
{}
Answer:
""".format(context, question)

response = qa_model(
    prompt,
    max_new_tokens=100
)

answer = response[0]["generated_text"]

# -----
# 10. Display results
# -----

print("\n-----")
print("RAG QUESTION ANSWERING SYSTEM")
print("-----")

print("\nRetrieved Context:")
print(context)

print("\nQuestion:")
print(question)

print("\nGenerated Answer:")
print(answer)

print("\nRAG process completed successfully.")
```

The terminal output shows the execution of the code, including the retrieval of context, the generation of an answer, and the final output of the RAG process. The output is as follows:

```
Device set to use cpu
Creating document embeddings...
Document stored successfully.

Enter your question: What is machine learning?

RAG QUESTION ANSWERING SYSTEM

Retrieved Context:
Machine Learning is a part of Artificial Intelligence. It allows computers to learn patterns from data and make predictions or decisions without being explicitly programmed. Deep Learning is a type of Machine Learning that uses artificial neural networks with multiple layers.

Question:
What is machine learning?

Generated Answer:
a part of Artificial Intelligence

RAG process completed successfully.
```